- col index - reset
- Titanic data set

$x \rightarrow$ move

$\downarrow$ y

features set

(Chess board)

$\boxed{y} = f(x)$

?

expert

data (M2)

$\hat{y} = P(y=1) = \dfrac{1}{1+e^{-(\theta_0+\theta_1 x_1+\theta_2 x_2}}$

$\uparrow$

y - actual

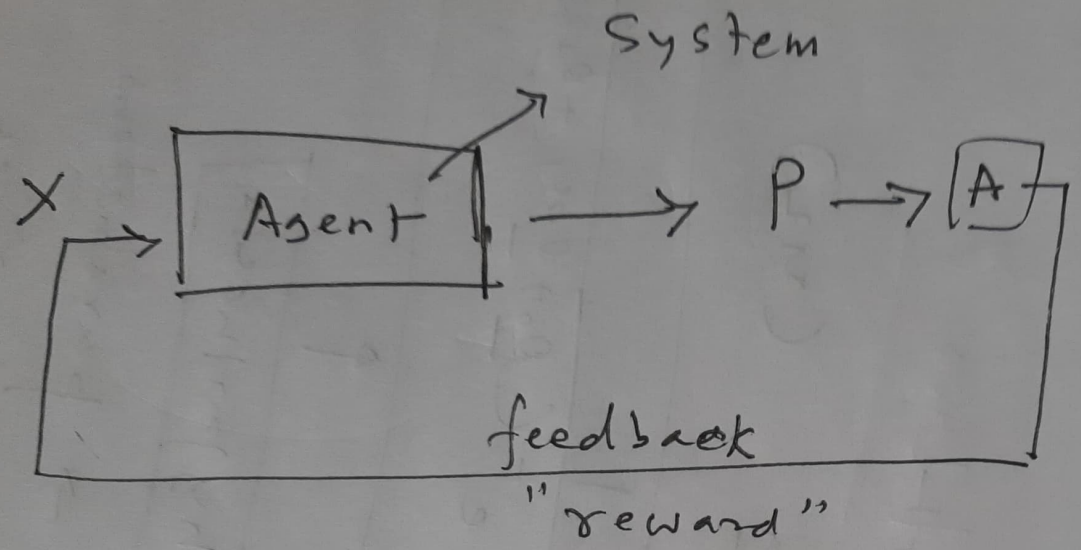
$\uparrow$

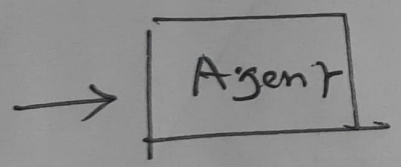
hypothesis

Logistic Regression

$\theta_0, \theta_1, \theta_2 \ldots \rightarrow$ parameters

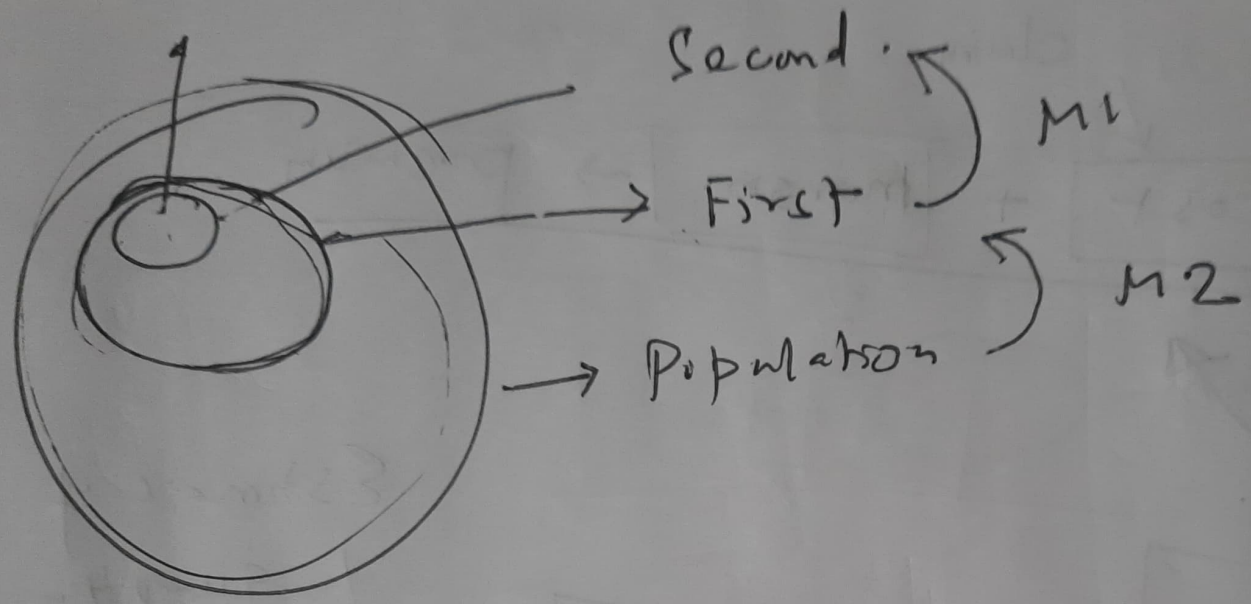
$x_1, x_2, \ldots \rightarrow$ features





$$\hat{y} = h_\theta(x) \rightarrow A1$$

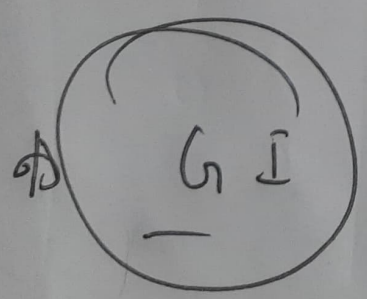
$$\hat{y}' = h'_\theta(x) \rightarrow A2$$



Distribution features

"Narrow Task"

"Generalized AI"



- GA

claim

$x \rightarrow \boxed{\text{cost}} + \boxed{\text{margin}} \rightarrow \text{premium}$

Estimate of quantities

Policy Bazar

10

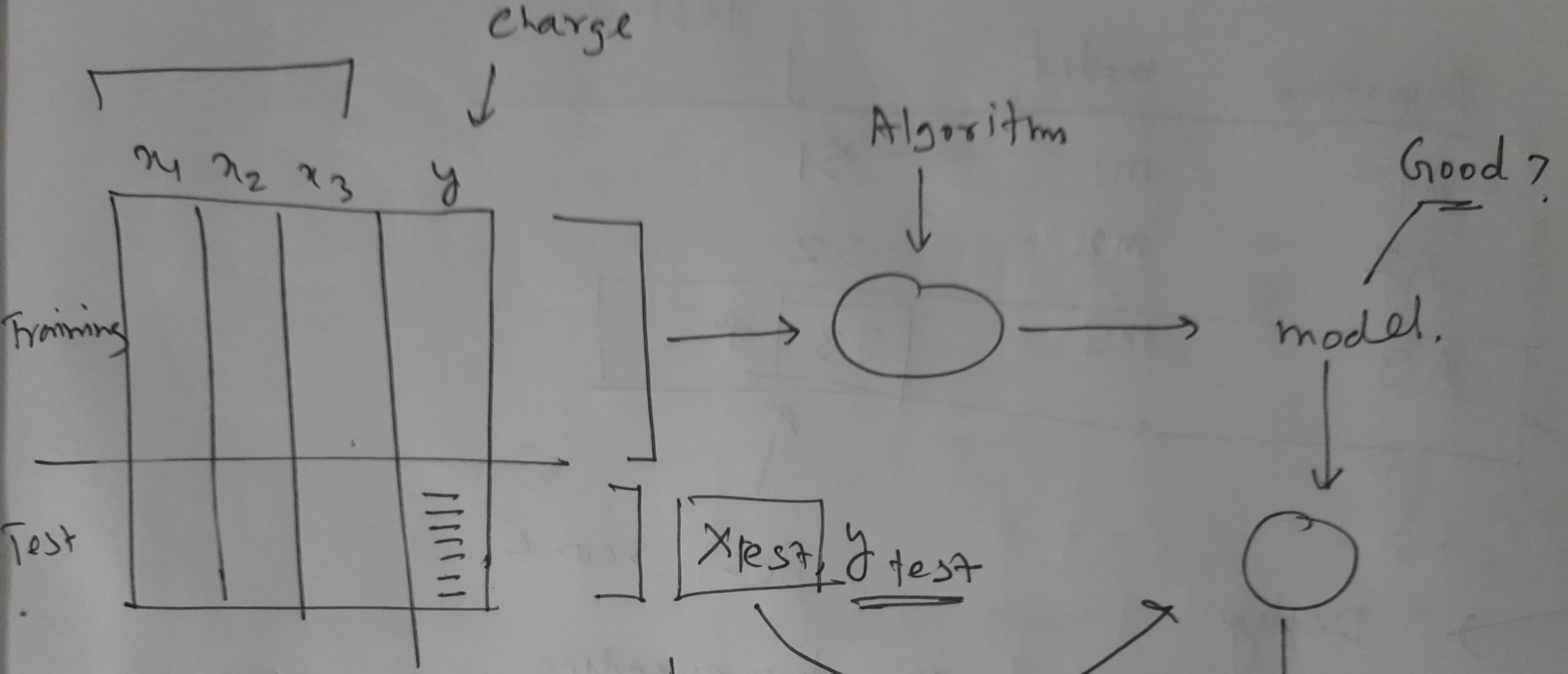
100

A → 10 → q_1 → q_1'

B → q_2 → q_2'

C → q_3 → q_3'

D → q_4



Training data

actual prediction

$$\text{Perf}(y_{test}, \hat{y}_{test})$$

$\hat{y}_{test}$ = estimate

$$\text{error} = (\hat{y}_{test} - y_{test}) \rightarrow \text{MSE}$$

$$\text{total error} = \frac{1}{m} \sum_{i=1}^{m} \left(\hat{y}_{test}^{(i)} - y_{test}^{(i)}\right)^2$$

test scores

Algorithm	model	
A1	m1	S1
A2	m2	S2
A3	m3	$\boxed{S3}$ *
⋮	⋮	

Deploy

Best score

Categorical One hot encoding.

age	region
30	R1
31	R2
⋮	R3
	⋮

Region R1	Region R2
1	0
0	1
0	0

OHE

Gender	M	F
M	1	0
F	0	1
null	0	1

X

~~MSE~~

SSE = $\sum_{i=1}^{m} \left(\hat{y}^{(i)} - y^{(i)} \right)^2$

Sum of squared error

SST = $\sum_{i=1}^{m} \left(\bar{y} - y^{(i)} \right)^2$

Sum of squared total

Standard Scaling

For every feature x,

$$Z = \frac{x - \bar{x}}{S_x}$$

$\bar{x}$ ← mean

S_x ← std deviation

= Z-score

Expected

SSE < SST

$$R^2 = 1 - \frac{SSE}{SST}$$

$0 < R^2 \leq 1$

$R^2 \leq 1$

R^2 ← Diagnosis

MSE

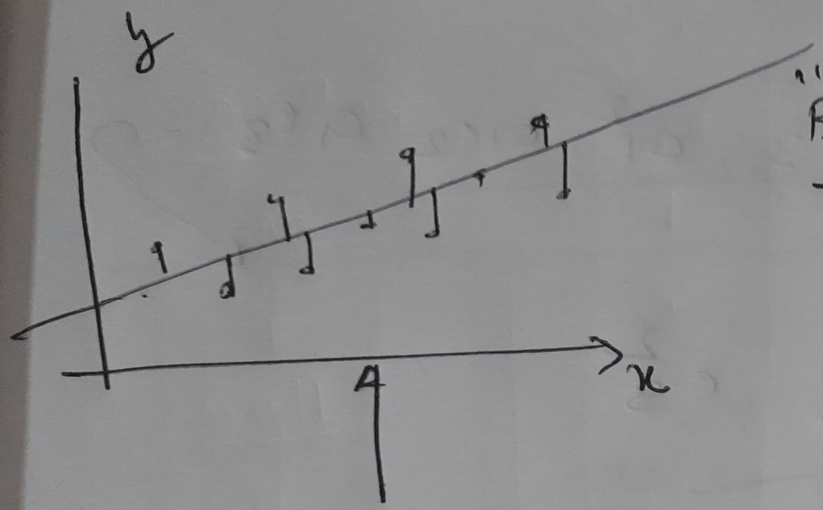
RMSE ← model selection

$$MAE = \frac{1}{m} \sum_{i=1}^{m} \left| \hat{y}^{(i)} - y^{(i)} \right|$$

→ less influenced by "high errors"

↓

so outliers



"Best fit" $\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots$

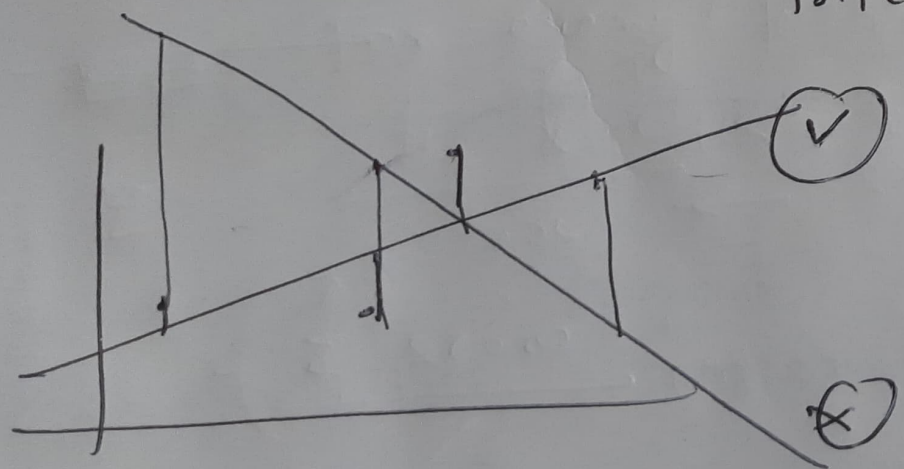
$+ \theta_j x_j$

j = no of features

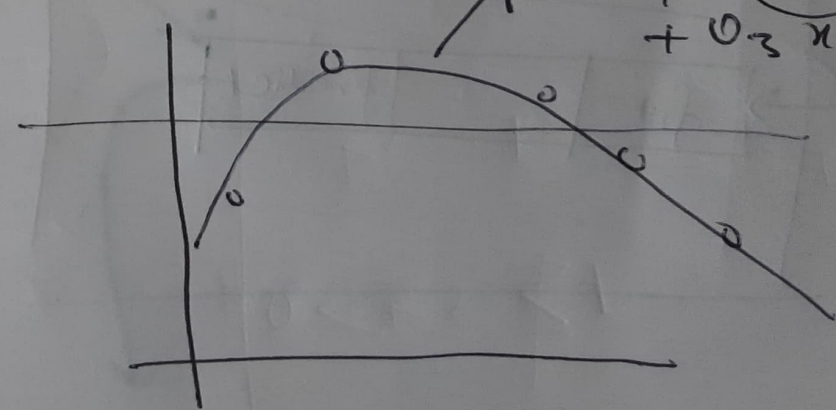
Hypothesis.

$$\hat{y} = h_\theta(x)$$
$$= a + bx$$
$$= \theta_0 + \theta_1 x_1$$

intercept, parameter

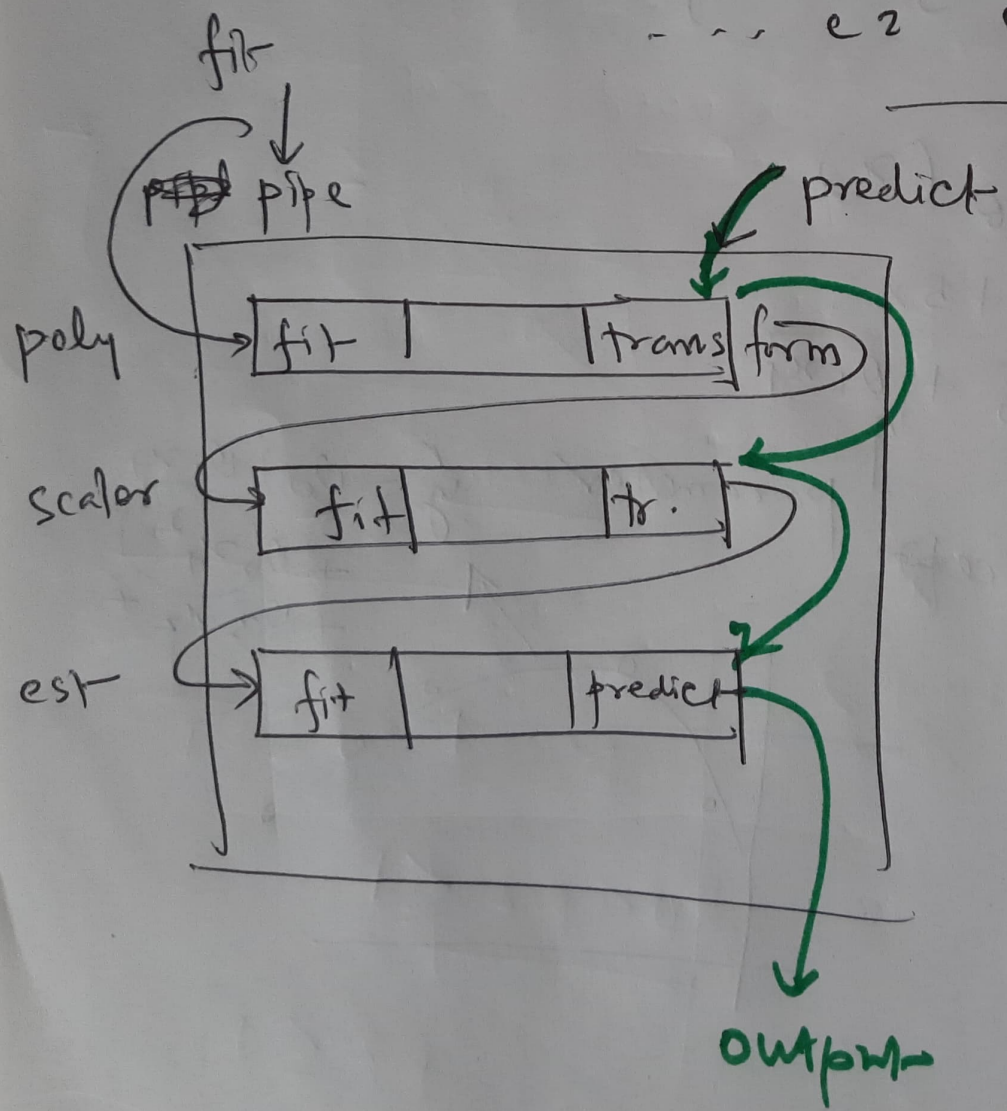


$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_2 + \cdots$

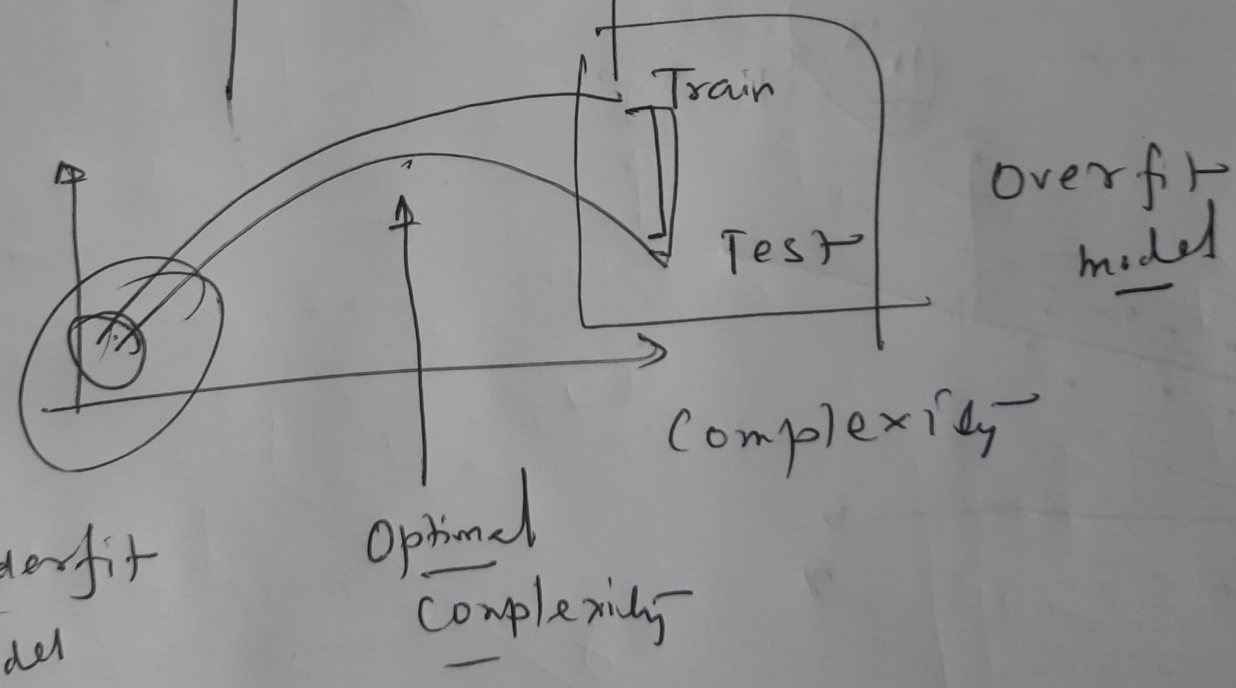


$c_1 \quad c_2 \quad c_3 \xrightarrow{\text{Poly}} c_1 \quad c_2 \quad c_3 \quad c_1^2 \quad c_1c_2 \quad c_1c_3 \ldots$

$\ldots \quad c_2^2 \quad c_2c_3 \quad c_3^2$



Degree	R2 (train)	R2 test
1	0.74	0.76
2	0.84	0.85
3	0.87	0.83
4	0.92	-0.19



$y = 5x + 6$

$y = x + 4$

$y = 4$